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September 13, 1976
SUPERSEDING
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September 11, 1967

FEDERAL SPECIFICATION

PIPE, CAST, GRAY AND DUCTILE IRON, PRESSURE (FOR WATER AND OTHER LIQUIDS)

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cast iron (gray iron and ductile iron) pressure pipe for conveying water, sewage, and other liquids.

1.2 Classification.

1.2.1 Types. Pipe shall be of the following types, as specified (see 6.2):

Type II - Push-on joint.
Type III - Mechanical joint.

1.2.2 Grades. Pipe shall be of the following grades, as specified (see 6.2):

Grade A - Gray iron 18/40 (18,000 pounds per square inch (psi) burst strength/40,000 psi modulus of rupture).
Grade B - Gray iron 21/45(21,000 psi burst strength/45,000 psi modulus of rupture).
Grade C - Ductile iron 60-42-10 (60,000 psi tensile strength/42,000 psi yield strength/10 percent elongation).

FSC 4710

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1.2.3 Thickness classes. Pipe shall be of thickness classes 50 through 56 in accordance with ANSI A21.50 for grade C pipe and thickness classes 20 through 30 in accordance with ANSI A21.1 for grades A and B pipe. Pipe shall be of thickness class as specified (see 3.5 and 6.2).

1.2.4 Sizes. Pipe shall be of nominal diameters from 3 to 54 inches in the following increments, as specified (see 6.2):

4 to 20 inches - 2-inch increments
24 to 54 inches - 6-inch increments

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agenices)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC, 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specification and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI) Standards:

A21.1 - Thickness Design of Cast-Iron Pipe
A21.4 - Cement-Mortar Lining for Cast-Iron and Ductile-Iron Pipe and Fittings for Water.
A21.5 - Polyethylene Encasement for Gray and Ductile Cast Iron Piping for Water and other Liquids.
A21.6 - Cast-Iron Pipe Centrifugally Cast in Metal Molds for Water or other Liquids.
A21.8 - Cast-Iron Pipe Centrifugally Cast in Sand-Lined Molds for Water or other Liquids.
A21.10 - Gray Iron and Ductile Iron Fittings, 2 inch through 48 inch for Water and other Liquids.
A21.11 - Rubber-Gasket Joints for Cast-Iron and Ductile-Iron Pressure Pipe and Fittings.
A21.15 - Flanged Cast-Iron and Ductile-Iron Pipe with Threaded Flanges.
A21.50 - Thickness Design of Ductile-Iron Pipe
A21.51 - Ductile-Iron Pipe, Centrifugally Cast in Metal Molds or Sand-Lined Molds for Water or other Liquids.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard commercial product. The pipe shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements. Additional or better features which are a part of the manufacturer's standard commercial product shall be included in the pipe being furnished. A standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.2 Material. All material shall be new and unused and as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. All materials shall be free from defects which would adversely affect the performance or maintainability of individual component or the overall assembly.

3.3 Description

3.3.1 Type I pipe. Type I pipe (calked joint) is not permitted by this specification.

3.3.2 Type II pipe. Type II pipe shall be designed for assembly by prepositioning a continuous, molded, rubber-ring gasket in an annular recess in the pipe socket and forcing the spigot end of the entering pipe into the socket, thereby compressing the gasket radially to the pipe to form a positive seal. The design and shape of the gasket and the annular recess shall be such that the gasket is locked in place against displacement as the joint is assembled. The joint design shall be in accordance with the manufacturer's standard practice. The manufacturer shall have qualified the design of the joint in accordance with ANSI A21.11. The spigot end of the pipe shall be suitably beveled to facilitate assembly. Type II pipe shall be centrifugally cast either in a metal mold or a sand-lined mold with use of grade A, B, or C iron.

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The specific class of type II pipe required shall conform to the composition, dimensions, tolerances, and mechanical properties as specified in ANSI A21.1, A21.6, A21.8, or A21.51, as applicable. When specified (see 6.2), and prior to the award of the contract, the manufacturer shall furnish the procuring activity drawings of the joint and gasket.

3.3.2.1 Type II sealing gasket and lubricant. Type II sealing gasket and lubricant shall be as specified in ANSI A21.11. The size and shape of the gasket shall be such as to provide adequate compressive force between the spigot and the socket after assembly to effect a positive seal under all combinations of joint and gasket tolerances.

3.3.3 Type III pipe. Type III pipe shall be furnished with a mechanical joint of the stuffing-box type, having a sealing gasket, a separate follower gland, and tee-head bolts with nuts used for joint assembly. Type III pipe shall be centrifugally cast in either a metal mold or a sand-lined mold with the use of grade A, B, or C iron. The specific class of type III pipe required shall conform to the composition, dimensions, tolerances, and mechanical properties as specified in ANSI A21.1, A21.6, A21.8 and A21.51, as applicable.

3.3.3.1 Type III sealing gasket. Type III sealing gasket shall be in accordance with ANSI A21.11.

3.3.3.2 Type III glands, bolts and nuts. Type III glands, bolts and nuts shall be in accordance with ANSI A21.11.

3.4 Strength. The minimum strength requirements for grades A and B pipe shall be in accordance with ANSI A21.6 and A21.8, except that for grade B pipe the bursting tensile strength shall be a minimum of 21,000 psi and the ring modulus of rupture shall be a minimum of 45,000 psi. The minimum strength requirements for grade C pipe shall be in accordance with ANSI A21.51.

3.5 Pipe thickness. Pipe thickness required shall be specified by thickness class (see 1.2.3 and 6.2), and shall be in accordance with the tables for standard thickness classes as given in ANSI A21.1 for grades A and B iron, and ANSI A21.50 for grade C. iron.

3.5.1 Wall thickness. Wall thickness shall be in accordance with the applicable ANSI standards and tables as specified in 3.5 for the thickness class specified.

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3.5.2 Working pressure. Working pressure shall be in accordance with the ANSI standards and tables as listed in Table I for the applicable thickness class specified.

TABLE I

Iron	ANSI Std.	Table Title
Grade A	A21.1	Schedule of barrel thickness for water pipe of 18/40 iron strength.
Grade B	A21.1	Schedule of barrel thickness for water pipe of 21/45 iron strength.
Grade C	A21.51	Standard thickness selection table for ductile-iron pipe.

3.5.3 Laying condition. Unless otherwise specified (see 6.2), pipe shall be furnished for installation with 5 feet of cover under laying condition B. Laying conditions shall be as defined in Table II.

TABLE II

Laying condition	Description
A	Pipe laid on flat-bottom trench, backfill not tamped.
B	Pipe laid on flat-bottom trench, backfill tamped.
C	Pipe bedded in gravel or sand, backfill tamped.

3.6 Laying length. The total laying length of pipe as installed shall be as specified (see 6.2). Nominal laying lengths of individual pipes shall be 18 or 20 feet, and as given in the applicable ANSI standards, depending on the method of casting. The nominal laying length shall be measured from the shoulder of the socket to the spigot end. Tolerance shall be as specified in ANSI A21.6 and A21.8 for grades A and B pipe, and ANSI A21.51 for grade C pipe.

3.7 Weight. The pipe weights shall be as specified in ANSI A21.6, A21.8, and A21.51 as applicable.

3.8 Cement-mortar lining. When specified (see 6.2 and 6.3), pipe shall be lined with cement mortar. The cement-mortar lining shall be in accordance with ANSI A21.4, and shall be standard thickness unless otherwise specified (see 6.2).

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3.9 Coating. When specified (see 6.2), pipe shall be coated inside and outside with a bituminous coating conforming to ANSI A21.6, A21.8, and A21.51. Cement mortar-lined pipe shall be furnished for potable water systems in accordance with ANSI A21.4. Bituminous interior coating shall be specified for all nonpotable systems. Polyethylene encasement during installation is permitted in accordance with ANSI A21.5.

3.9.1 Coating on glands. When specified (see 6.2), glands shall be coated with a bituminous coating conforming to ANSI A21.11.

3.10 Identification marking. Each pipe shall be identified with markings in accordance with ANSI A21.6 and A21.8 for grades A and B pipe, and ANSI A21.51 for grade C pipe. In addition to the markings specified above, grade A pipe shall have the letters "GR A", and grade B pipe shall have the letters "GR B" and "21/45" clearly marked on it. In addition to the above markings, type II pipe shall be marked in accordance with ANSI A21.11.

3.11 Recommended method of assembly. When specified (see 6.2), complete instructions for type II joint assembly shall be furnished by the manufacturer.

3.12 Workmanship. Workmanship shall be in accordance with the best commercial practice for this type of product. All pipe shall be clean and sound without defects which will impair their service. Repairing of defects by welding or other method shall not be allowed if such repairs will affect the serviceability of the pipe or its capability to meet the strength requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for examination and testing. Sampling for examination and testing of pipe, sealing gaskets, glands, bolts, nuts, and lubricant shall be in accordance with the manufacturer's standard practice and the requirements of ANSI A21.6, A21.8, A21.11, and A21.51 as applicable.

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4.3 Tests

4.3.1 Pipe tests. Pipe shall be subject to the tests listed in Table III in accordance with the standard specified.

4.3.2 Rubber gasket test. Rubber gasket material for type II and type III pipe shall be tested as specified in ANSI A21.11.

4.3.3 Lubricant test. The lubricant used for lubricating the parts of type II joints for assembly shall be as specified in ANSI A21.11.

4.3.4 Glands, nuts, and bolts test. Glands, nuts, and bolts shall be tested as specified in ANSI A21.11.

TABLE III. Tests Required.

	Grades A & B (Gray cast iron pipe)		Grade C (Ductile-iron pipe)
Property	Cast in metal molds	Cast in sand-lined molds	Cast in metal or sand-lined molds
Standard	ANSI A21.6	ANSI A21.8	ANSI A21.51
Material composition	Chemical analysis	Chemical analysis	Not applicable
Resistance to leakage	Hydrostatic test	Hydrostatic test	Hydrostatic test
Tensile Strength	Full Length Burst	Full Length Burst	Tensile test
Min yield strength	Not applicable	Not applicable	Tensile test
Min elongation	Not applicable	Not applicable	Tensile test
Modulus of rupture	Talbot-Strip and ring tests	Talbot-strip and ring tests	Not applicable
Modulus of elasticity	Talbot-strip test	Talbot-strip test	Not applicable
Breaking load and deflection	Not applicable	Test-bar test	Not applicable
Hardness	Rockwell hardness test	Not applicable	Not applicable
Impact strength	Not applicable	Not applicable	Impact test

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4.3.5 Cement-mortar lining test. The cement-mortar lining materials and the thickness of the lining shall be tested as specified in ANSI A21.4.

4.4 Test data. All test data shall be recorded, and shall be available to the Government on request.

4.5 Examination. Each pipe, gasket, gland, bolt, and nut and the lubricant shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Preparation for delivery inspection. The packing and marking of the pipe and joint materials shall be inspected to verify conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Pipe. Unless otherwise specified (see 6.2), pipe under 24 inches in diameter of like kind and size shall be bundled by placing the first layer on 4-inch by 4-inch timbers. Nominal size lumber shall be used between the layers. The number of pipes per layer and the number of layers per bundle shall be the contractor's standard practice. Appropriate chocks shall be secured on the end of the timbers to prevent movement of the pipe and to facilitate strapping. The strapping shall conform to QQ-S-781, class 1, type II, heavy duty finish B. The straps and separators shall be placed one-sixth the length of the bundle from each end, with intermediate straps and separators placed equidistant but not exceeding 48 inches on centers. Pipe 24 inches in diameter and greater shall be shipped loose.

5.1.1.2 Joint material (bolts, nuts, gaskets, glands, and joint lubricant). Joint material shall be packed in wood boxes in accordance with PPP-B-601, overseas type; or PPP-B-621, class 2. Contents shall be cushioned and braced to prevent movement. The boxes shall be closed

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and strapped in accordance with the appendix to the box specification. Strapping shall be class I, type I, heavy duty, finish B in accordance with QQ-S-781

5.1.2 Level B.

5.1.2.1 Pipe. The pipe shall be packed as specified for level A, except the strapping need not be zinc coated.

5.1.2.2 Joint material (bolts, nuts, gaskets, glands, and joint lubricant). The joint material shall be packed as specified for level A, except that the boxes shall conform to PPP-B-621, class 1; or PPP-B-601, domestic type, and the strapping need not be zinc coated.

5.1.3 Level C. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract, pipe shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract, pipe shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Types II and III pipe are for use in conveying water, sewage, or other liquids, and are used to advantage in wet trenches, where flexibility is required or where excessive expansion or contraction may occur.

6.1.1 Pipe grades. Gray iron pipe (grades A and B) is suitable for most routine applications, but is becoming obsolete. Ductile iron pipe (grade C) is a superior material and should be specified for reasons of economy and durability.

6.1.2 Temperature limitations. The maximum service temperature will be limited by the rubber gaskets. Where an application with liquid temperatures in excess of 180° Fahrenheit may exist, the manufacturer should be consulted.

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6.1.3 Method of specifying pipe thickness. Pipe wall thickness may be specified by thickness class or standard wall thickness.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and nominal diameter of pipe (see 1.2 and 6.1.1).
(c) When drawings of gasket and joint are required for type II pipe (see 3.3.2).
(d) Pipe thickness class or wall thickness (see 3.5 and 6.1.3).
(e) Laying condition (see 3.5.3).
(f) Total laying length (length installed) required (see 3.6)
(g) When polyethylene encasement is required (see 3.9).
(h) When bituminous coating is required on pipe and glands (see 3.9 and 3.9.1).
(i) When cement-mortar lining is required and lining thickness if other than standard (see 3.8 and 6.3).
(j) When complete assembly instructions are required for type II pipe (see 3.11).
(k) Level of packing required (see 5.1).
(l) When pipe is to be packed other than as specified (see 5.1.1.1).

6.3 Lining. Pipe for water service shall be provided with a standard lining in accordance with ANSI A21.4. When special conditions require, linings can be furnished in thicknesses equal to twice the standard thickness specified in ANSI A21.4.

6.4 Fittings. Fittings for use with pipe covered by this specification are covered by ANSI A21.10 entitled "Gray-Iron and Ductile-Iron Fittings, 2 inch through 48 inch for Water and Other Liquids", and ANSI A21.15 entitled "Flanged Cast iron and Ductile-iron Pipe with Threaded Flanges".

6.5 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provision of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered

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in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9 (n) are not invoked the data shall be delivered in accordance with the contract requirements.

Military Custodians:

Army - ME
Navy - YD
Air Force - 11

Civil Agency Coordinating Activity:

GSA-FSS

Preparing activity:

Navy - YD

Project No. 4710-0309

Review activity:

Army - CE

User activities:

Army - MR
Navy - CE, MC

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